

## Full Length Article

# Beyond semantics: Exploiting propagation structures with dual-adapter LLMs for fake news detection

Aojie Si , Shizhan Chen, Xiaobao Wang, Yueheng Sun\*, Jiaai Guo 

College of Intelligence and Computing, Tianjin University, No. 92 Weijin Road, Tianjin, 300072, China

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## ABSTRACT

The rapid spread of fake news across multiple platforms and domains poses significant risks to public opinion and societal stability. Traditional approaches to fake news detection, which predominantly rely on content analysis, are becoming less effective as large language models (LLMs) lower the cost of generating fake news. It is necessary to explore more robust, generalized detection methods that can handle complex real-world situations. This paper proposes **Propagation Structure-Augmented LLMs (PSALLM)**, a novel fake news detection model that leverages LLMs to understand and perceive news propagation structures. Specifically, we design two complementary adapters within a pre-trained LLM model: an adapter with non-weight-decay (**ANWD**) to capture invariant structural features across domains, and an adapter with weight-decay (**AWD**) to capture domain-specific structural shifts. We further introduce an adaptive distillation strategy based on domain similarity to dynamically balance the influence of both adapters. Extensive experiments on four real-world cross-domain datasets demonstrate that PSALLM consistently outperforms state-of-the-art baselines, confirming its effectiveness in handling diverse news domains and manipulation strategies. The code will be published after the end of the anonymity period.

## 1. Introduction

With the continuous development of the information society, the volume of fake news is growing exponentially and spreading rapidly on online platforms. The spread of fake news not only greatly misleads public opinion and damages the online environment, but also poses a serious threat to social stability and development. Therefore, how to timely and accurately determine the authenticity of news has become an important issue that urgently needs to be addressed.

The traditional methods for detecting fake news mainly focus on analyzing the semantics (Wu et al., 2020), style (Monti et al., 2019), and emotions of news content (Min et al., 2022), and gradually introducing the social attributes of news, such as news release media, user interaction indicators, and the publishing preferences of news authors. While these approaches have demonstrated moderate success, they are typically evaluated under closed-world assumptions, where training and test data share the same underlying distribution (Peng et al., 2025, 2024). The advent of LLMs has dramatically lowered the barrier to generating highly convincing fake news (Madine, 2024), enabling its rapid proliferation across diverse platforms and topical domains. As a result, models trained on static, in-distribution data struggle to generalize in real-world settings. This shift underscores the pressing need for more generalized

and robust fake news detection frameworks capable of operating effectively under open-world, cross-domain conditions.

Recent research has demonstrated that models leveraging news propagation structures provide greater robustness and stability compared to those that rely solely on content features or user profiles (Sun et al., 2023). Propagation patterns, such as repost trees or user interaction graphs, are less susceptible to superficial manipulations and often reflect deeper behavioral and relational cues inherent in how misinformation spreads. These structural features are generally domain-invariant, making them well-suited for cross-domain fake news detection (Han et al., 2020). At the same time, Large Language Models (LLMs) have achieved impressive generalization capabilities across various natural language understanding tasks, including cross-domain inference (Leng & Xiong, 2024). However, due to their training primarily on flat textual input, LLMs lack an explicit mechanism for modeling structural dependencies in relational data, such as the temporal and topological characteristics of news diffusion (Gong et al., 2023). This observation reveals a key insight. Propagation structures offer stable, manipulation-resistant features, while LLMs provide strong semantic reasoning and generalization. Relying solely on either results in limited performance under real-world conditions, especially when detecting fake news across diverse domains or adversarial scenarios. **Therefore, it is essential to**

\* Corresponding author.

E-mail addresses: 2023248038@tju.edu.cn (A. Si), shizhan@tju.edu.cn (S. Chen), wangxiaobao@tju.edu.cn (X. Wang), yhs@tju.edu.cn (Y. Sun), 3022244365@tju.edu.cn (J. Guo).

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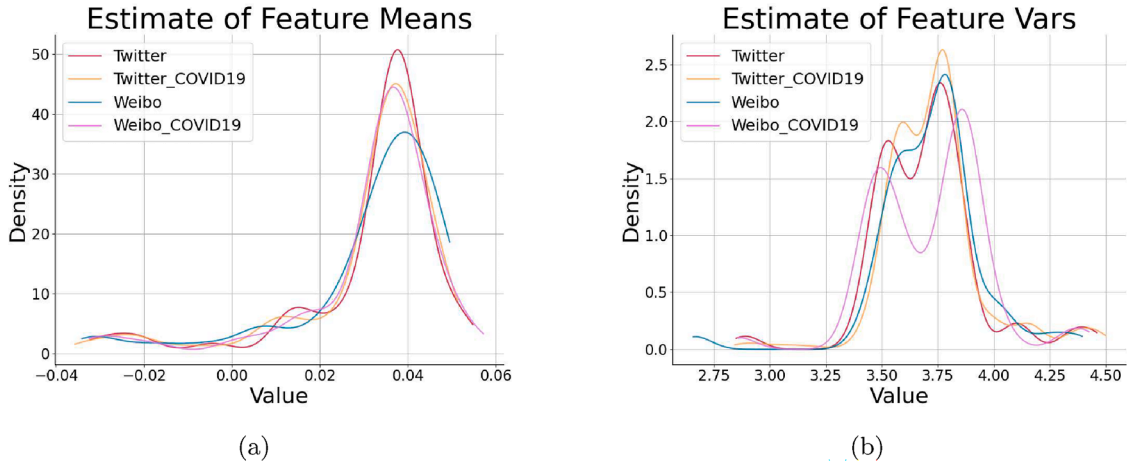


Fig. 1. The distributional differences across four datasets. Domain similarity is computed using Earth-Mover Distance (EMD) (Cui et al., 2018; Oh et al., 2022).

**integrate propagation structure modeling with LLMs into a unified framework.** This integration enables the model to leverage both semantic expressiveness and structural awareness, leading to more robust and generalizable fake news detection.

Previous studies have established that data from different domains typically exhibit distinct distributions in feature space (Peng et al., 2025). Motivated by this, we construct feature distribution maps that incorporate propagation structure information across multiple domain-specific datasets, as illustrated in Fig. 1. However, our empirical analysis reveals a more nuanced reality: although propagation structures demonstrate a degree of cross-domain stability (Phan et al., 2023), they exhibit subtle yet non-negligible domain-specific biases. As shown in Fig. 1(a), the average propagation patterns across domains remain similar, indicating structural consistency. However, Fig. 1(b) illustrates that the variance in these structures differs significantly across domains, revealing subtle but impactful inter-domain variations. These findings present a dual challenge for leveraging LLMs in fake news detection. On one hand, capturing the stable structural patterns is essential for ensuring cross-domain generalization. On the other hand, adapting to domain-specific deviations is critical for avoiding overgeneralization and performance degradation. Addressing both aspects simultaneously within a single model remains an open problem.

To tackle this challenge, we propose **PSALLM**, a Propagation Structure-Augmented Large Language Model for fake news detection. PSALLM incorporates two complementary adapters into a pre-trained LLaMA backbone (Touvron et al., 2023): a **Non-Weight Decay Adapter (ANWD)** that captures invariant structural features across domains, and a **Weight Decay Adapter (AWD)** that models domain-specific structural shifts. The two adapters are trained progressively and coordinated via a domain similarity-aware adaptive distillation mechanism, which dynamically adjusts their contributions based on structural alignment. Moreover, to rigorously evaluate the model's generalization capability, we adopt a zero-shot evaluation protocol, in which the support set contains no samples or domain-specific information from the target domains.

Our contributions are summarized as follows:

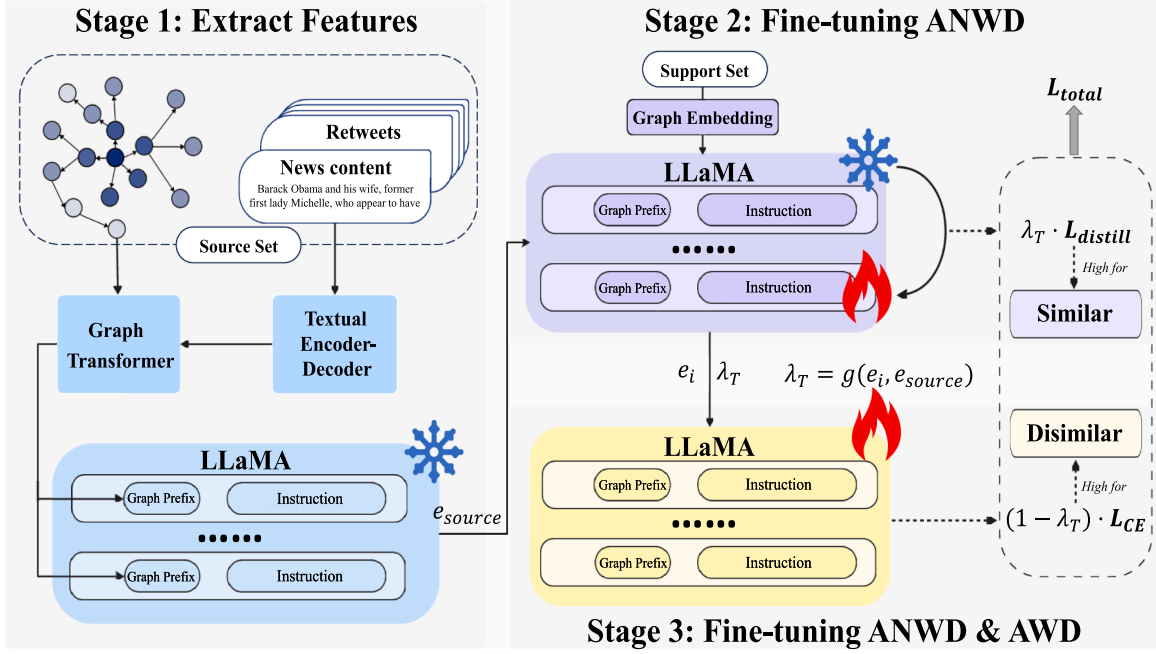
- We introduce **PSALLM**, the first propagation-aware LLM framework designed for robust cross-domain fake news detection.
- We propose a dual-adapter strategy to decouple stable and domain-sensitive propagation structure representations and design a domain similarity-guided adaptive distillation method to dynamically balance the contributions of the two adapters.

- We conduct extensive experiments on four real-world cross-domain datasets, where **PSALLM** consistently outperforms state-of-the-art baselines.

## 2. Related works

Traditional fake news detection methods focused mainly on linguistic features extracted from news content. Subsequently, more sophisticated models incorporated social-contextual features to enhance detection performance. For example, TriFN (Shu et al., 2019) jointly models the relationship between publishers and news, as well as the interactions between users and news articles. SMAN (Yuan et al., 2020) integrates news content with credibility scores of both publishers and users. FR-Detect framework (Jarrahi & Safari, 2021) combines content-related features with publisher activeness and influence metrics. UPFD (Dou et al., 2021) utilizes users' historical posts to construct preference embeddings, which are combined with news content to form a propagation graph used by a GNN for classification. EHIAN (Wu et al., 2021) extracts evidence from related news articles to support fact-checking. In addition to text, fake news detection methods introduce multimodal features. For example, GE-NSLM (He et al., 2025) reveals deceptive patterns in text and images. BMR (Ying et al., 2023) captures complementary fake news signals from text, image modality, and image semantics.

News propagation structure has received increasing attention due to its ability to capture how news spreads through social networks. Graph-based methods such as GCN and GAT have shown strong performance by modeling repost and comment behaviors as graph structures (Lin et al., 2021; Meel & Vishwakarma, 2021). Bi-GCN (Bian et al., 2020) enhances this capability by constructing bi-directional rumor propagation graphs. GLAN (Yuan et al., 2019) effectively captures the relationships among source tweets, retweets, and users as a heterogeneous graph. UPFD (Dou et al., 2021) introduces external conditions through users' historical posts, combined with propagation structures, to effectively identify fake news. Hy-DeFake (Su et al., 2025) captures high-order correlations between news and users to learn distinctive embeddings for fake news detection. GANM (Chang et al., 2024) strengthens the capture and integration of global information and key information from news propagation structures. PSGT (Zhu et al., 2024) excels at capturing long-range propagation dependencies, while HG-SL (Sun et al., 2023) leverages global propagation behaviors for structural learning. GACL (Sun et al., 2022) focuses on reducing noise in propagation graphs, and GAMC (Yin et al., 2024) explores potential structures in unlabeled datasets via unsupervised graph mining. These findings confirm that propagation



**Fig. 2.** The network architecture of PSALLM. If the propagation structure of the source set is similar to that of the support set, ANWD is selected as the primary adapter. Otherwise, AWD is employed as the primary adapter.

patterns offer domain-invariant cues essential for robust cross-domain detection.

With the emergence of large language models (LLMs) such as GPT (Achiam et al., 2023) and LLaMA (Touvron et al., 2023), several recent studies have explored their application in fake news detection. For example, Su et al. (2023) conduct adversarial training by rewriting real news with LLMs to improve robustness. Li et al. (2024) propose a multi-round retrieval-augmented LLM approach to enhance detection performance via evidence retrieval. LEKD (Chen et al., 2025) combines the strengths of SLMs, external knowledge, and LLMs to enhance text-centric fake news detection. Subsequent studies (Pavlyshenko, 2023) show that fine-tuning LLMs on fake news corpora can significantly improve textual detection ability. However, LLMs trained solely on text often lack robustness in real-world scenarios due to a lack of propagation structural information.

To bridge this gap, our proposed PSALLM introduces a unified framework that integrates a graph-enhanced prefix into an LLM and employs two progressive adapters (ANWD to AWD) to balance intra-domain and inter-domain generalization. A domain-adaptive distillation strategy further enables dynamic adapter selection based on structural similarity, providing an effective solution for robust and generalized fake news detection in real-world settings.

### 3. Problem statement

We formulate the cross-domain fake news detection task as a supervised classification problem under a domain shift setting. Let  $\mathcal{D} = \{(C_i, G_i, y_i)\}_{i=1}^N$  denote a dataset of  $N$  news events. For each event  $i$ :

- $C_i$  represents the semantic content including news and retweet text.
- $G_i = (\mathcal{V}_i, \mathcal{E}_i)$  denotes the propagation structure graph, where nodes  $\mathcal{V}_i$  represent users or posts, and edges  $\mathcal{E}_i$  represent the forwarding relationship.
- $y_i \in \{0, 1\}$  is the ground-truth label (Real or Fake).

To bridge the gap between structural graphs and the LLM's input space, we define a graph-enhanced prefix  $P_i = \phi(G_i, C_i)$ , which captures the fused structural and semantic features of event  $i$ .

We consider dividing two datasets from different sources into two distinct domains, the source domain  $\mathcal{D}_S$  and the target domain  $\mathcal{D}_T$ . During the model training phase, the model is trained purely on  $\mathcal{D}_S$ , while during the model evaluation phase, the model is evaluated on  $\mathcal{D}_T$ . Let  $e_{\text{source}}$  denote the normalized aggregate embedding of the entire source domain  $\mathcal{D}_S$ , serving as a global representation of its domain-specific features. For a given training batch  $m$  sampled from the support set  $S \subset \mathcal{D}_S$ , let  $e_m$  denote the batch-wise hidden representation that jointly encodes the graph-enhanced prefixes  $P_i$  and instruction prompts  $I_i$  derived from the  $m$ -th batch.

During the progressive fine-tuning phase, the optimization is dynamically guided by a domain-adaptive coefficient  $\lambda_T = g(e_m, e_{\text{source}})$ . This coefficient estimates the domain shift between the current batch representation  $e_m$  and the global source embedding  $e_{\text{source}}$ , thereby adaptively modulating the contributions of the two adapters to enhance the model's generalization across domains. The goal is to learn a mapping function  $f_\theta(C, G) \rightarrow \hat{y}$  that minimizes the prediction error on the target domain  $\mathcal{D}_T$ .

## 4. Methodology

In this section, we propose a fine-tuned LLM framework for fake news detection based on news propagation structures, as illustrated in Fig. 2. The proposed model leverages graph prefix to enable the LLM to directly comprehend the propagation structures, without requiring textual descriptions of the propagation graph (Guo et al., 2023). We first extract features from the multimodal graph-text structure (§4.1), and then progressively train two adapters (Yang et al., 2024) by exploiting the modular nature of adapter tuning (§4.2). Finally, we apply an adaptive distillation strategy between ANWD and AWD to balance their respective contributions (§4.3).

### 4.1. Source feature extraction

In the first stage of our framework, we leverage the representational capacity of LLMs to deterministically model the feature distribution of the source set during pre-training (Zhang et al., 2024). This is achieved by extracting structured information from the source

set-including graph-augmented prefixes and task instructions-and integrating it into the LLM's input representation. As a result, the model internalizes the global structural and semantic characteristics of the source set, which are subsequently preserved and effectively utilized when fine-tuning lightweight adapters for downstream cross-domain tasks.

#### 4.1.1. Graph-enhanced prefix tuning.

We adopt the effective methodology proposed in GraphLLM (Chai et al., 2023). Specifically, we first employ a textual Transformer encoder-decoder to extract semantic representations  $C_i$  from the news propagation nodes. These semantic features are then fused with the structural representation  $G_i$ , obtained via a Graph Transformer that encodes the news propagation graph, to construct a graph-enhanced prefix  $P_i$ . Finally, this graph-enhanced prefix  $P_i$  is integrated into the LLM, thereby enhancing the model's ability to understand the propagation structure.

#### 4.1.2. Batch feature extraction

To better capture the structural and semantic consistency between the source set and the support set, we introduce a batch-wise feature extraction mechanism. For each input instance, both the graph-enhanced prefix  $P_i$  and its associated instruction prompt  $I_i$  are fed into the model to generate hidden representations.

Let  $M$  denote the number of batches, and let  $e_m$  be the output embedding of the  $m$ -th batch. This embedding, produced by the model, jointly encodes the local propagation pattern  $P_i$  and the instruction prompt  $I_i$  derived from the  $m$ -th batch. Importantly, the attention mechanism employed here is shared with the adapter-based fine-tuning stage, allowing for alignment between upstream and downstream structural modeling. This consistency facilitates a smooth transfer of domain-aware representations.

#### 4.1.3. Batch feature fusion

While batch-level representations are effective for local context modeling, each batch only captures a partial view of the overall propagation and content distribution. To generate a unified and domain-representative embedding, we introduce a normalized feature fusion mechanism. First, we compute a raw aggregate embedding of all batches:

$$e_{\text{total}} = \sum_{m=1}^M e_m, \quad (1)$$

where  $e_m \in \mathbb{R}^{B \times L \times d}$  denotes the  $m$ -th batch embedding,  $B$  is the batch size,  $L$  the sequence length, and  $d$  is the hidden dimension. Then, we apply Root Mean Square (RMS) (Zhang & Sennrich, 2019) normalization, inspired by its successful application in transformer-based architectures and its compatibility with LLaMA-2 architecture (Touvron et al., 2023):

$$e_{\text{source}} = \omega \odot \frac{e_{\text{total}}^{(b,\ell)}}{\sqrt{\frac{1}{d} \sum_{k=1}^d (e_{\text{total}}^{(b,\ell,k)})^2 + \epsilon}}, \quad (2)$$

where  $b \in [1, B]$ ,  $\ell \in [1, L]$ ,  $\omega \in \mathbb{R}^d$  is a learnable scaling vector, and  $\epsilon$  ensures numerical stability. The resulting  $e_{\text{source}} \in \mathbb{R}^{B \times L \times d}$  serves as the teacher signal in our adaptive distillation framework, capturing both semantic information and structural patterns across the full source set.

### 4.2. Adapter fine-tuning

#### 4.2.1. Design of adapters

We divide the trainable adapter parameters of the pretrained LLM equally into two adapters: ANWD and AWD. In both adapters, model fine-tuning is performed using the Adam optimizer (Diederik, 2014), which enables efficient parameter updates with minimal computational cost. Each adapter is capable of learning both the semantic and structural information of news propagation.

However, the two adapters exhibit different capacities in feature learning. ANWD, without weight decay, is better suited to distinguish invariant structural features across domains. In contrast, AWD, with weight decay applied, improves generalization and is more effective at capturing domain-specific structural shifts, as illustrated in Fig. 3.

#### 4.2.2. Progressive learning of two adapters

Although we have designed two adapters tailored for different functions, in practice, using two independently functioning adapters is insufficient for capturing the overall news propagation structures. This limitation is further validated in our ablation studies (§5.4). Therefore, an adaptive mechanism is needed to fully leverage both adapters by dynamically selecting the more suitable one based on the domain similarity between the input news and the source domain news. In such a mechanism, the primary adapter is selected to maximize its advantage of capturing potential characteristics across domains, while the other serves as auxiliary support.

In the first stage, we obtain source feature  $e_{\text{source}}$  that integrate both propagation structure and semantic information through feature extraction and fusion. This source feature is essential for guiding the adaptive selection of adapters.

In the second stage, similar to the feature extraction process in Stage 1, we construct the embedding  $e_m$  that combines the graph-enhanced prefix  $P_i$  with instruction prompt  $I_i$  derived from the  $m$ -th batch, and  $e_m$  is fed into the ANWD adapter. Crucially, the parameters of ANWD remain frozen during this phase. Let  $x \in \mathbb{R}^{s \times v}$ , where  $s$  is the number of tokens and  $v$  is the vocabulary size. Let  $y \in \{0, 1, \dots, v-1\}^{s \times v}$  denote the ground-truth labels, and let  $\hat{p}_i = \text{Softmax}(x_i)$  represent the predicted probability distribution for the  $i$ -th token. The cross-entropy loss is defined as:

$$\mathcal{L}_{CE} = -\frac{1}{s} \sum_{i=1}^s \log \hat{p}_i[y_i]. \quad (3)$$

In the third stage, using the previously obtained  $e_m$ , we proceed to the synchronous update of both adapters. Simultaneously, we perform knowledge distillation from the teacher representation  $e_{\text{source}}$ , modulated by a domain similarity-based adaptive coefficient  $\lambda_T$ . It is derived from a function  $g(e_m, e_{\text{source}})$ , which provides an estimate of the domain shift in ANWD, enabling the model to adaptively select the appropriate adapter based on the similarity across domains. The relevant calculation of  $g(e_m, e_{\text{source}})$  is described in detail later (§4.3).

Since cosine similarity is employed for classification, the distillation loss is also defined based on cosine similarity:

$$\cos = \frac{1}{B \cdot L} \sum_{b=1}^B \sum_{\ell=1}^L \frac{e_{\text{source}}^{(b,\ell)} \cdot e_m^{(b,\ell)}}{\|e_{\text{source}}^{(b,\ell)}\|_2 \cdot \|e_m^{(b,\ell)}\|_2}, \quad (4)$$

Accordingly, the final loss function is defined as:

$$\mathcal{L}_{\text{total}} = (1 - \lambda_T) \cdot \mathcal{L}_{CE} + \lambda_T \cdot \mathcal{L}_{\text{distill}}, \quad (5)$$

where  $\mathcal{L}_{\text{distill}} = 1 - \cos$ . When the domain similarity is high,  $\lambda_T$  is large, and the method increases the distillation strength, thereby enhancing the model's ability to capture intrinsic commonalities in propagation structures. In this case, the model relies more on ANWD for adaptation, as activating AWD would reduce its alignment with the teacher representation  $e_{\text{source}}$ . Conversely, when the domain similarity is low,  $\lambda_T$  is small, leading to reduced distillation intensity. This shifts the primary adaptation role to AWD, allowing the model to better capture domain-specific structural shifts.

#### 4.3. Domain adaptive coefficient for distillation

The adaptive coefficient  $\lambda_T$  controls the activation strength of the adapter modules and plays a crucial role in the two proposed adapter training strategies. We propose a domain similarity-based computation



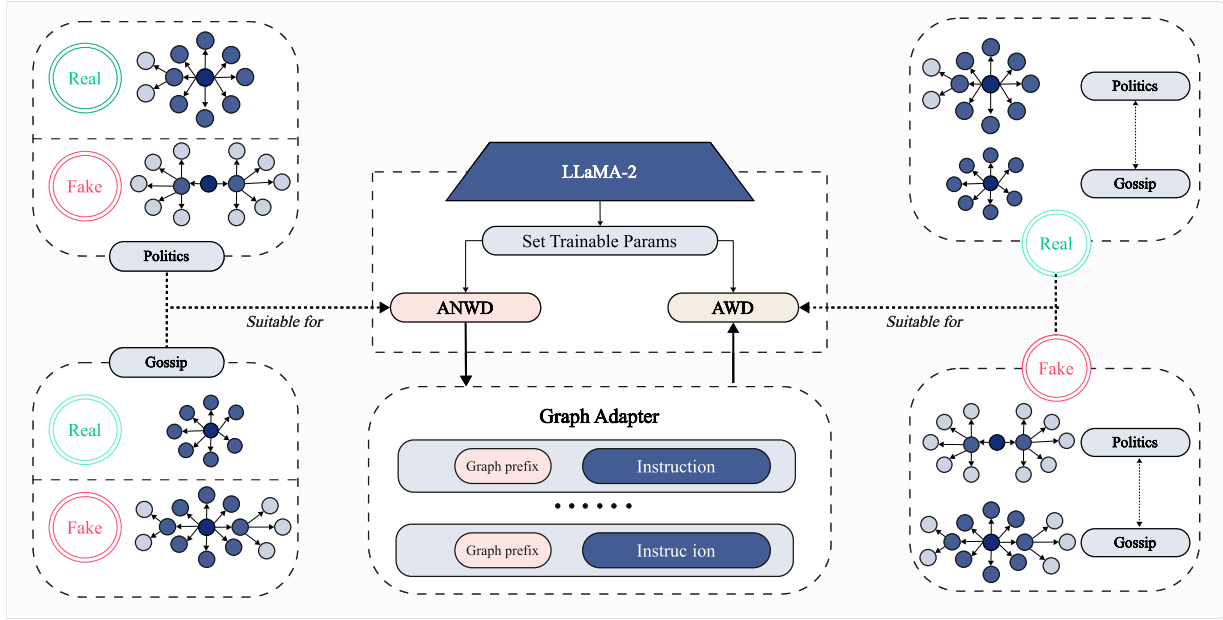


Fig. 3. Progressive training of ANWD and AWD, along with their respective domain applicability.

method for  $\lambda_T$ , which selectively utilizes two key statistical representations,  $e_{\text{source}}$  and  $e_m$ , to efficiently guide the distillation process without requiring additional supervision. This adaptive distillation method optimizes the synergy between the two training strategies. Both the source representation  $e_{\text{source}} \in \mathbb{R}^{B \times L \times d}$  and the current training batch embedding  $e_m \in \mathbb{R}^{B \times L \times d}$  are sequences of token-level features.

To estimate cross-domain alignment, we introduce two complementary similarity scores derived from  $e_{\text{source}}$  and  $e_m$ : one operating at the fine-grained token level, and the other at a coarse-grained instance level. The discrepancy between these two scores reflects the stability of feature correspondence across abstraction levels, which we use to modulate distillation strength.

#### 4.3.1. Fine-grained alignment score

The fine-grained alignment score captures low-level structural consistency in propagation sequences across domains. Specifically, we compute the cosine similarity between corresponding token-level embeddings from the source domain representation  $e_{\text{source}}$  and the current training batch embedding  $e_m$ , thereby measuring point-wise feature correspondence along the entire propagation path:

$$S_{\text{fine}} = \sum_{b=1}^B \sum_{\ell=1}^L \cos(e_{\text{source}}^{(b,\ell)}, e_m^{(b,\ell)}). \quad (6)$$

#### 4.3.2. Coarse-grained alignment score

The coarse-grained alignment score, in contrast, reflects high-level semantic alignment by abstracting away token-level variations. To obtain this score, we first aggregate features within each instance by averaging over its constituent tokens to form a compact instance-level representation. We then compute the cosine similarity between the resulting aggregated representations from the source domain  $e_{\text{source}}$  and the current training domain  $e_m$ , emphasizing consistency at the semantic or instance level rather than at individual token positions:

$$S_{\text{coarse}} = \sum_{b=1}^B \cos\left(\frac{1}{L} \sum_{\ell=1}^L e_{\text{source}}^{(b,\ell)}, \frac{1}{L} \sum_{\ell=1}^L e_m^{(b,\ell)}\right), \quad (7)$$

#### 4.3.3. Adaptive coefficient

The final domain-adaptive distillation coefficient  $\lambda_T$  is derived from the absolute difference between fine-grained and coarse-grained similarity scores. First, we normalize both scores to account for their different

scales:

$$\bar{S}_{\text{fine}} = \frac{1}{N_f} S_{\text{fine}}, \quad \bar{S}_{\text{coarse}} = \frac{1}{N_c} S_{\text{coarse}}, \quad (8)$$

where  $N_f = B \cdot L$  is the total number of token pairs, and  $N_c = B$  is the number of instances.

We then compute the absolute difference between the normalized scores:

$$\Delta = |\bar{S}_{\text{fine}} - \bar{S}_{\text{coarse}}|, \quad (9)$$

where a smaller value of  $\Delta$  indicates higher cross-domain similarity, implying that the current training domain is more structurally and semantically aligned with the source domain.

Finally, we map  $\Delta$  to an adaptive coefficient  $\lambda_T \in (0, 1)$  using the function:

$$\lambda_T = \exp(-\beta \cdot \Delta), \quad (10)$$

where  $\beta$  is a scaling factor that controls the sensitivity of  $\lambda_T$  to domain discrepancy, as analyzed in detail later (§5.5). A larger  $\lambda_T$  indicates higher similarity between the source and current training domains, leading to stronger distillation. In practice, we observe that on simpler domains such as Twitter,  $\lambda_T$  typically ranges in  $[0.4, 0.9]$ , whereas on more complex and heterogeneous domains like Weibo, it falls within  $[0.2, 0.9]$ , reflecting the varying degrees of domain shift. This result confirms that our mechanism can adaptively adjust distillation intensity based on domain shift.

## 5. Experiments

### 5.1. Datasets

To evaluate the effectiveness of our proposed model, we conduct experiments on six real-world rumor detection datasets, as shown in Table 1. Due to the complexity and scale of LLMs, we sample only 50 retweets and comment nodes for each news event. To verify the model's capability in capturing news propagation structures, we construct four cross-domain dataset settings:

**Group 1:** Twitter15 (Ma et al., 2018) as the training set and Twitter16 (Ma et al., 2018) as the test set.

**Group 2:** Twitter16 as the training set and Twitter15 as the test set.

**Table 1**  
Statistics of the cross-domain datasets.

| Statistic     | Twitter15 | Twitter16 | Twitter | Twitter_COVID19 | Weibo | Weibo_COVID19 |
|---------------|-----------|-----------|---------|-----------------|-------|---------------|
| Source Tweets | 1490      | 818       | 1154    | 400             | 4649  | 399           |
| Rumor         | 1116      | 613       | 575     | 252             | 2313  | 253           |
| Non-Rumor     | 374       | 205       | 579     | 148             | 2336  | 146           |
| Avg Retweets  | 25        | 23        | 30      | 46              | 49    | 26            |

**Table 2**  
Performance of four cross-domain datasets with node count limited to 50.

| Methods           | Twitter15 $\rightarrow$ Twitter16 |              | Twitter16 $\rightarrow$ Twitter15 |              | Twitter $\rightarrow$ Twitter_COVID19 |              | Weibo $\rightarrow$ Weibo_COVID19 |              |
|-------------------|-----------------------------------|--------------|-----------------------------------|--------------|---------------------------------------|--------------|-----------------------------------|--------------|
|                   | Acc.                              | $F_1$ .      | Acc.                              | $F_1$ .      | Acc.                                  | $F_1$ .      | Acc.                              | $F_1$ .      |
| LSTM              | 58.33                             | 59.75        | 54.25                             | 53.85        | 41.69                                 | 41.19        | 42.10                             | 43.61        |
| CNN               | 61.96                             | 58.96        | 55.33                             | 55.53        | 40.25                                 | 41.50        | 42.87                             | 43.11        |
| Rumor-GAN         | 61.30                             | 59.45        | 57.01                             | 57.02        | 43.96                                 | 43.84        | 43.23                             | 43.55        |
| PPA-WAE           | 71.25                             | 70.55        | 65.03                             | 65.71        | 45.28                                 | 45.98        | 53.28                             | 52.46        |
| GACL              | 69.37                             | 72.36        | 67.04                             | 66.73        | 51.62                                 | 51.78        | 53.77                             | 53.00        |
| GAMC              | 74.66                             | 73.32        | <u>74.12</u>                      | <u>73.31</u> | 54.13                                 | 54.58        | <u>59.23</u>                      | <u>61.63</u> |
| RAGAR             | 72.77                             | 71.52        | 73.49                             | 70.72        | 54.02                                 | 53.63        | 58.10                             | 56.98        |
| GraphLLM          | <u>76.87</u>                      | <u>76.32</u> | 72.68                             | 72.37        | <u>54.50</u>                          | <u>54.96</u> | 52.07                             | 51.15        |
| <b>ContentLLM</b> | 74.23                             | 68.15        | 65.17                             | 66.14        | 50.51                                 | 51.13        | 47.45                             | 42.44        |
| <b>PSALLM-2</b>   | 77.48                             | 77.18        | <b>75.52</b>                      | <b>73.95</b> | 59.25                                 | 59.43        | <b>69.61</b>                      | <b>68.01</b> |
| <b>PSALLM-3</b>   | <b>77.60</b>                      | <b>78.08</b> | 74.53                             | 73.48        | <b>60.50</b>                          | <b>60.99</b> | 64.28                             | 63.27        |
| $\uparrow$ (%)    | <b>0.95</b>                       | <b>2.31</b>  | <b>1.89</b>                       | <b>0.87</b>  | <b>11.01</b>                          | <b>10.97</b> | <b>17.52</b>                      | <b>10.35</b> |

**Group 3:** Twitter (Ma et al., 2017) as the training set and Twitter\_COVID19 (Lin et al., 2022) as the test set.

**Group 4:** Weibo (Ma et al., 2016a) as the training set and Weibo\_COVID19 (Lin et al., 2022) as the test set.

## 5.2. Baseline

To evaluate the training effectiveness of our model, we selected some models based on neural networks, a text-based LLM, and a graph network-based LLM:

**LSTM** (Ma et al., 2016b) is an LSTM-based rumor detection model to learn feature representations of retweets over time.

**CNN** (Yu et al., 2017) uses a CNN model for detecting rumors by framing the retweets as a fixed-length sequence.

**Rumor-GAN** (Ma et al., 2019) uses a generative adversarial network (GAN) in which a generator is designed to produce conflicting voices to pressure the discriminator to learn stronger rumor representations.

**PPA-WAE** (Zhang et al., 2021) utilizes a neural network model which is combined with a feedforward network on propagation trees to learn the semantics of the trees and their propagation structure.

**GACL** (Sun et al., 2022) utilizes a new graph adversarial contrastive learning method to capture the structural information of rumor propagation trees.

**GAMC** (Yin et al., 2024) uses the graph autoencoder with masking and contrastive learning for fake news detection.

**RAGAR** (Khaliq et al., 2024) proposes that CoRAG and ToRAG verify the facts of the news. Because the training process is not involved here, we only detect fake news in the news content of the target domain.

**GraphLLM** (Chai et al., 2023) enables LLM to understand the propagation structure of fake news through graph prefixes.

**ContentLLM** as a baseline, we modify GraphLLM by removing all modules related to propagation structure, which adopts the pre-trained LLaMA-2-7B model, allowing it to detect fake news purely based on textual information via fine-tuning.

**PSALLM-2** adopts the pre-trained LLaMA-2-7B model.

**PSALLM-3** adopts the pre-trained LLaMA-3-8B model.

For these cross-domain datasets, we evaluate Accuracy (Acc.) and the weighted F1-score ( $F_1$ .), which is the sample-size-weighted average of the per-class F1 scores.

### 5.2.1. Implementation details.

Each news propagation structure is limited to a maximum of 50 nodes, and the maximum input length for textual content is set to 512 tokens. For baseline methods, the maximum number of training epochs is set to 50. In contrast, due to the increased complexity of our model, we limit the training to 15 epochs to prevent overfitting.

For our model, we instantiate two variants of our model, PSALLM-2 and PSALLM-3, respectively based on the LLaMA-2-7B and LLaMA-3-8B model. Although these base models differ in architecture, our method remains compatible with both through simple input sequence truncation or padding. Importantly, the adapter modules are kept identical across both variants to ensure consistent adaptation behavior and a fair comparison. Both the instruction encoder and the task decoder are implemented as 4-layer Transformer blocks, following the architectural style of the underlying LLaMA models. The adapter configuration uses an adapter length of 5, an adapter hidden dimension of 768, and 4 attention heads. To model the propagation structure, we applied random walk sampling with a walk length of 5 to capture the local structural context of each node. The resulting subgraphs are encoded using an MLP with 3 hidden layers, serving as the structure encoder for the graph-enhanced prefix. All experiments are conducted on a single NVIDIA A800 GPU with a batch size of 16, using the Adam optimizer (Diederik, 2014) with an initial learning rate of  $5 \times 10^{-5}$  and a minimum learning rate of  $5 \times 10^{-6}$ .

### 5.3. Overall performance

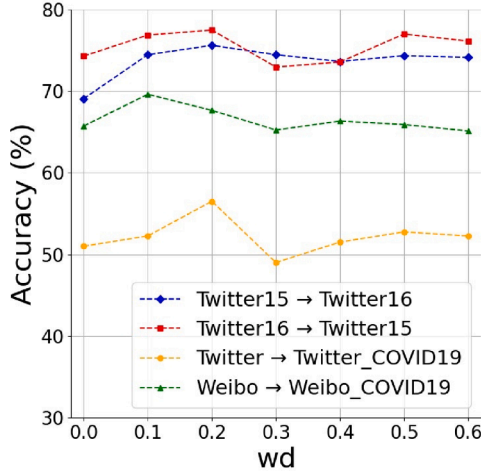
The results are presented in Table 2. It is evident that the results of the second group of experiments are consistently superior to those of the first group, demonstrating the effectiveness of capturing propagation structure information across cross-domain datasets. Among them, GAMC performs the best in the second group, exhibiting strong generalizability in cross-domain datasets.

In the third group of methods, we introduce LLMs. While RAGAR outperforms traditional news detection methods, its performance is inferior

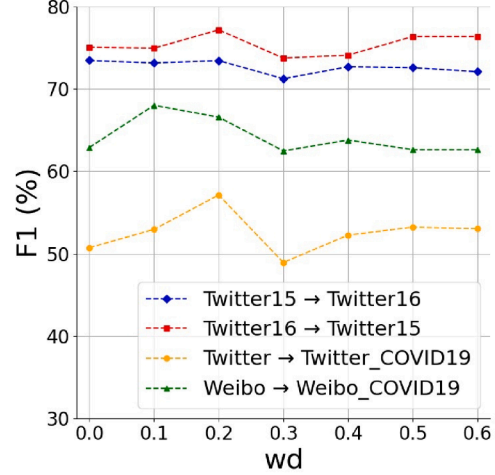
**Table 3**

Ablation study on the network design of our model on four cross-domain datasets.

| Methods     | Twitter15 $\rightarrow$ Twitter16 |              | Twitter16 $\rightarrow$ Twitter15 |              | Twitter $\rightarrow$ Twitter_COVID19 |              | Weibo $\rightarrow$ Weibo_COVID19 |              |
|-------------|-----------------------------------|--------------|-----------------------------------|--------------|---------------------------------------|--------------|-----------------------------------|--------------|
|             | Acc.                              | $F_1$ .      | Acc.                              | $F_1$ .      | Acc.                                  | $F_1$ .      | Acc.                              | $F_1$ .      |
| <b>Ours</b> | <b>77.48</b>                      | <b>77.18</b> | <b>75.52</b>                      | <b>73.95</b> | <b>59.25</b>                          | <b>59.43</b> | <b>69.61</b>                      | <b>68.01</b> |
| w/o Dis     | 76.49                             | 76.09        | 73.04                             | 72.92        | 53.25                                 | 53.88        | 45.52                             | 38.68        |
| w/o AC      | 76.74                             | 76.02        | 73.58                             | 73.71        | 50.25                                 | 50.34        | 44.04                             | 36.52        |
| w/o ANWD    | 76.38                             | 75.56        | 73.85                             | 69.82        | 55.75                                 | 55.59        | 63.01                             | 63.63        |
| w/o ACDiS   | 75.27                             | 75.24        | 73.25                             | 71.47        | 47.25                                 | 46.08        | 43.55                             | 35.47        |
| w/o ANWDDiS | 76.62                             | 75.20        | 72.84                             | 73.01        | 51.25                                 | 51.92        | 36.74                             | 22.09        |



(a) Accuracy



(b) F1-Score

**Fig. 4.** The influence of parameter  $wd$  on PSALLM across the four cross-domain datasets.

to models based on propagation structures. GraphLLM combines LLMs with news propagation structures and semantic information, yielding experimental results similar to GAMC. However, our model significantly outperforms other baseline models.

In particular, the accuracy of the Twitter to Twitter\_COVID19 and Weibo to Weibo\_COVID19 datasets significantly surpasses the performance of other models on these two cross domain datasets. Although it performs optimally on the Twitter15 to Twitter16 and Twitter16 to Twitter15 datasets, the improvement over baseline models is relatively modest. This is because the Twitter15 and Twitter16 datasets are nearly identical and have simpler propagation structures, with an average retweet node count of around 25. In contrast, significant improvements are observed in two other cross-domain datasets with larger domain shifts, where the average number of retweet nodes is higher than in the Twitter15 and Twitter16 datasets. The complexity of these datasets further demonstrates that our model is more suitable for detecting fake news in complex real-world scenarios.

We have also integrated the LLaMA-3-8B model into our framework. While its performance on certain datasets is marginally lower than that of the LLaMA-2-7B model, it still consistently outperforms the baseline models. This performance gap is probably due to mismatched parameters in the pre-trained model. However, these results underscore the robust transferability of our framework, highlighting its ability to adapt to different model architectures and still achieve superior performance compared to other baselines.

Through experimental analysis, we find that when the Weibo dataset, with an average of 49 retweets, is used as the training set, our model excels in effectively capturing the news propagation structure to the greatest extent, outperforming all other baseline models. This demonstrates that PSALLM is capable of fully learning the intricate characteristics of news propagation structures, highlighting its superior ability to understand and model the complexities of real-world news dissemination.

#### 5.4. Ablation study

To evaluate the effectiveness of each module in our model, we conduct comparative experiments by removing one or two components. The experimental settings are as follows: **1) w/o Dis**: Removes the distillation loss module from AWD. The model is trained using only the cross-entropy loss. **2) w/o AC**: Disables the adaptive coefficient  $\lambda_T$ , thus treating both adapters equally without domain similarity adjustment. **3) w/o ANWD**: Removes the ANWD adapter module and applies the same weight decay strategy as AWD. The distillation module and adaptive coefficient  $\lambda_T$  are retained. **4) w/o ACDiS**: Removes both the adaptive coefficient  $\lambda_T$  and the distillation loss module. Only the two adapter modules are retained. **5) w/o ANWDDiS**: Removes both the ANWD adapter and the distillation module, but keeps the adaptive coefficient  $\lambda_T$ . In this setting, both adapters are trained using cross-entropy loss.

The results are presented in Table 3. It is evident that all ablated variants perform worse than the full model, demonstrating that each component contributes meaningfully to the overall performance. By removing the ANWD module, it is evident that when only one adaptive distillation adapter module is retained, its impact on the overall model is minimal. However, on the cross-domain datasets Twitter to Twitter\_COVID19 and Weibo to Weibo\_COVID19, its performance is still significantly worse than that of the original model. When the ACDiS module is removed and only the two distinct adapter modules remain, the model's performance drops significantly on the cross-domain datasets, which further validates the importance of domain-adaptive distillation in our model. Additionally, by removing the Dis and AC modules, we find that the distillation loss module is more effective for model performance than the adaptive coefficient module. Finally, when the ANWDDiS module is removed, the model's performance on Weibo to Weibo\_COVID19 reaches its lowest point, underscoring that the distillation mechanism (Dis) plays a more critical role in the effectiveness of our original model."

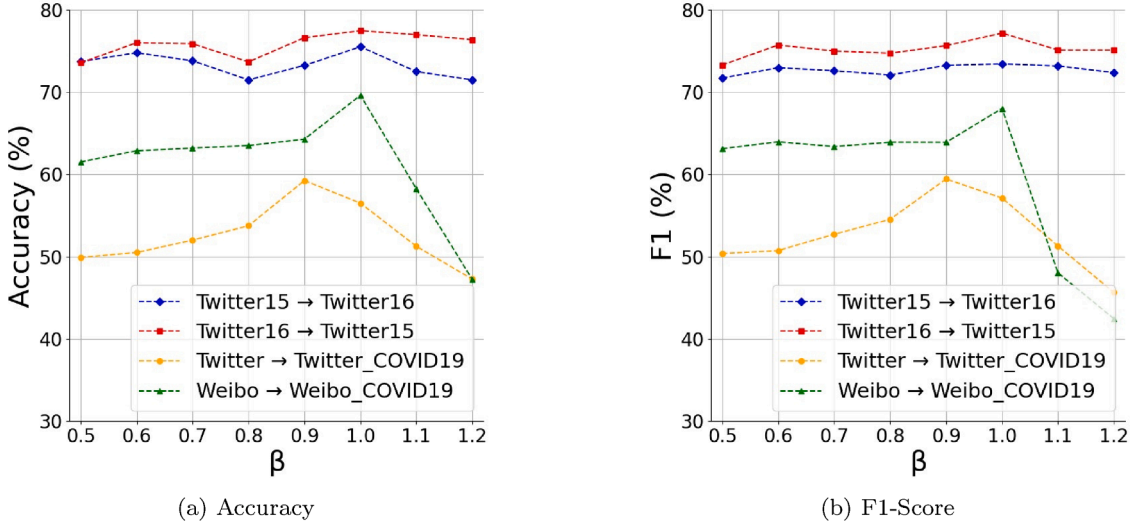


Fig. 5. The influence of parameter  $\beta$  on PSALLM across the four cross-domain datasets.

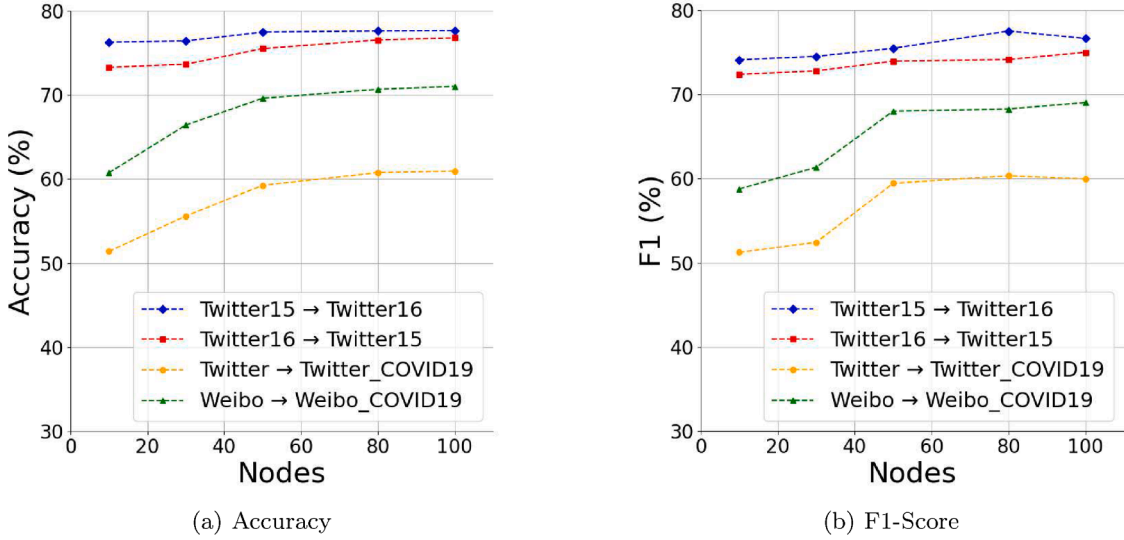


Fig. 6. Influence of node count on PSALLM across the four cross-domain datasets.

### 5.5. Parameter analysis

We further analyze the influence of two key hyperparameters: weight decay  $wd$  and scaling factor  $\beta$  on the performance of our model across the four cross-domain datasets. We first determine the optimal value for  $wd$ , with  $\beta = 1.0$  fixed by default, as shown in Fig. 4.

Through experimental analysis, we observe that smaller  $wd$  typically have a positive impact on model performance. The model's accuracy reaches its highest when  $wd$  is around 0.2. As  $wd$  continues to increase, the AWD adapter becomes over-regularized, leading to the model's inability to effectively capture the details of the propagation structure.

After identifying the best setting for  $wd$ , we then search for the optimal value  $\beta$  based on this configuration, as illustrated in Fig. 5. Experimental analysis shows that as  $\beta$  increases from 0.5 to around 1.0, the performance of the model generally improves. During this phase, the adjustments made by the adapters are not overly aggressive, and the differences between intra-domain and inter-domain similarities have a minimal impact. However, when  $\beta > 1.0$ , the model performance decreases with the increasing value of  $\beta$ . At this point, the model focuses more on capturing inter-domain differences in propagation structures, which

in turn reduces its ability to capture the stable relationships within the propagation structures.

### 5.6. Nodes analysis

In previous experiments, we selected only the top 50 retweets for each news article. In this section, we utilize the LLaMA-2-7B model and explore different node selections to further validate PSALLM's capability to capture news propagation structures, as shown in Fig. 6. We observe that as the number of nodes increases, PSALLM's detection ability improves, with the rate of improvement slowing after reaching 50 nodes. The slowdown in performance is due to the insufficient length of the graph-enhanced prefix embedded in the large model, limiting its ability to capture additional structural features. This demonstrates that our model can strongly detect fake news in real-world scenarios, even when the propagation structure is limited to a maximum of 50 nodes. These results highlight the model's robustness and adaptability, enabling it to maintain strong performance even when the news propagation process is constrained.



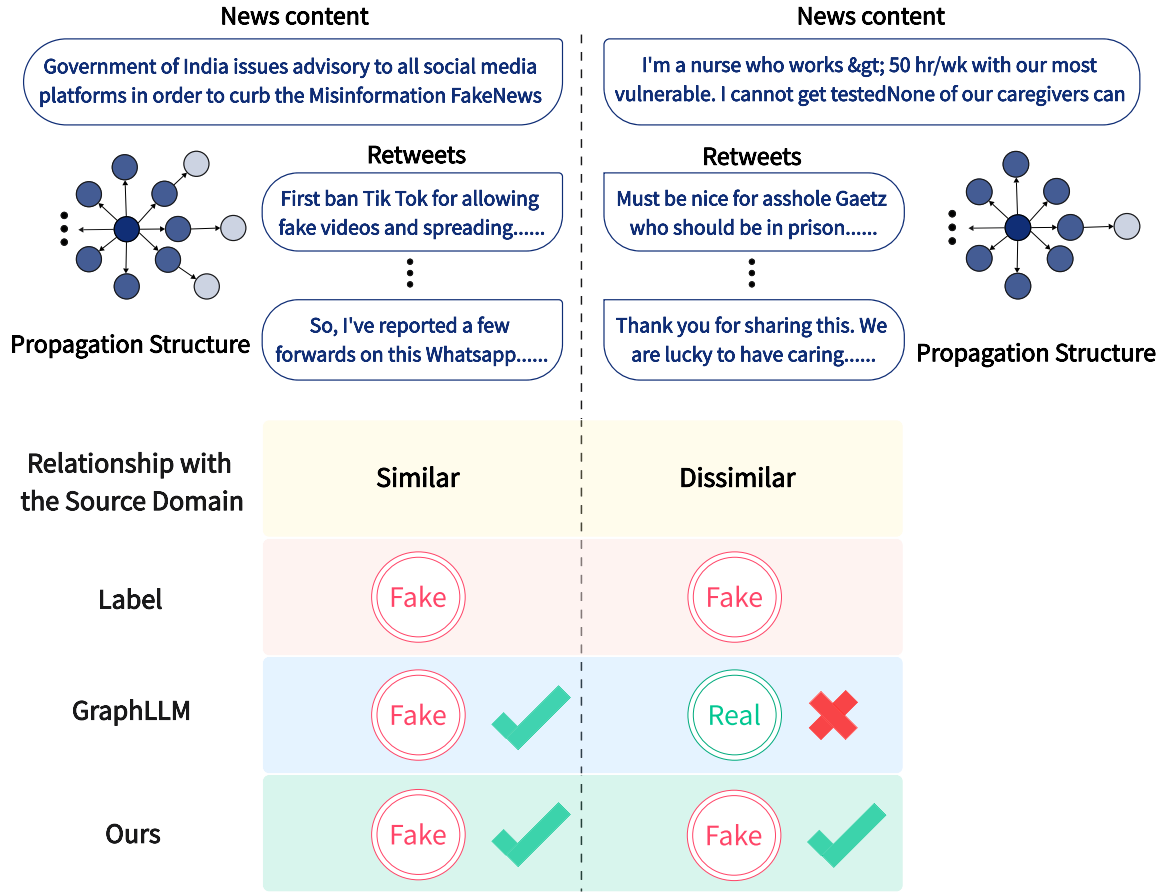


Fig. 7. Case studies between GraphLLM and ours.

### 5.7. Case study

As illustrated in Fig. 7, we present a comparative case study using two representative test samples drawn from the Twitter to Twitter\_COVID19 domain adaptation scenario to further analyze the behavior of our model. In the first case, where the target news item exhibits a propagation structure highly similar to patterns seen in the source set, both GraphLLM and our proposed PSALLM successfully classify the instance. This result aligns with expectations, as structural consistency facilitates reliable generalization across models. In contrast, the second case involves a news item whose propagation structure diverges significantly from those observed in the source domain. In this scenario, GraphLLM misclassifies the sample, whereas PSALLM correctly identifies it, demonstrating its superior identification capabilities.

We attribute this to the synergy between our dual-adaptor design and the domain-adaptive distillation strategy, which together allow the model to disentangle domain-specific structural shifts while maintaining semantic coherence. This indicates that our model can effectively capture subtle variations in propagation structures and semantics caused by domain differences. Furthermore, this underscores our model's robust ability to discern news propagation structures.

## 6. Conclusion

In this paper, we introduce PSALLM, a framework that enhances LLMs with interpretable propagation structure modeling for fake news detection. By integrating both structural and semantic cues into a graph-enhanced prefix, PSALLM enables the model to capture the dynamics of news propagation more effectively. To further improve domain generalization, we incorporate a dual-adaptor architecture and

a domain-adaptive distillation strategy. Extensive experiments on four real-world cross-domain datasets, along with comprehensive ablation studies, demonstrate the robustness and effectiveness of our approach.

This work underscores the potential of enhancing LLMs through propagation structure-augmented fine-tuning of dual-adapters, and opens up a promising direction for integrating LLMs with structural modeling of news propagation.

### CRedit authorship contribution statement

**Aojie Si:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation; **Shizhan Chen:** Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization; **Xiaobao Wang:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization; **Yueheng Sun:** Visualization, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization; **Jiaai Guo:** Validation, Software, Data curation.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Supplementary material

Supplementary material associated with this article can be found in the online version at [10.1016/j.neunet.2026.108904](https://doi.org/10.1016/j.neunet.2026.108904).

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